EXERCISE 3

MOSHE MATABANE

DATA ANALYTICS

BRIGHTLEARN

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```sql
--Question 01: Classify each product by price into the three tiers below
SELECT product_name, price,
    CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
        WHEN price < 100 THEN 'Budget'
    END AS price_category
    FROM exercise.class.products;
```

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product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

```sql
--Question 02: Label each order by its value
SELECT customer_name,
      amount,
    CASE
         WHEN amount >= 1000 THEN 'High Value'
         WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
         WHEN amount < 500 THEN 'Low Value'
    END AS order_value_category
    FROM exercise.class.orders;
```

Add parameter

Table

	customer_name	amount	order_value_category
1	Alice	150	Low Value
2	Bob	560	Medium Value
3	Charlie	999	Medium Value
4	Diana	45	Low Value
5	Ethan	1200	High Value

```sql
--Question 3: Categorize each employee's position level using both department and salary
SELECT emp_name, department, salary,
    CASE
       WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
       WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
       ELSE 'Staff'
   END AS position_level
   FROM exercise.class.employees;
```

Add parameter

Table

	emp_name	department	salary	position_level
1	John	IT	85000	Senior IT
2	Sara	HR	60000	Experienced HR
3	Mark	IT	75000	Staff
4	Lucy	Finance	95000	Staff
5	Tom	HR	55000	Staff

```sql
--Question 4: Assign each student a letter grade based on their score
SELECT student_name, score,
    CASE
        WHEN score >= 90 THEN 'A'
        WHEN score BETWEEN 80 AND 89 THEN 'B'
        WHEN score BETWEEN 70 AND 79 THEN 'C'
        WHEN score BETWEEN 60 AND 69 THEN 'D'
        WHEN score < 60 THEN 'F'
    END AS grade
    FROM exercise.class.students;
```

Add parameter

Table

	student_name	score	grade
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ella	68	D

```sql
--Question 05: Lable delivery performance based on the time taken
SELECT delivery_id, delivery_time_minutes,
    CASE
        WHEN delivery_time_minutes <= 30 THEN 'Fast'
        WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
        ELSE 'Late'
END AS performance
FROM exercise.class.deliveries;
```

Add parameter

Table

	delivery_id	delivery_time_minutes	performance
1	1	45	On Time
2	2	80	Late
3	3	30	Fast
4	4	65	Late
5	5	100	Late

```sql
--Question 06: Convert the numeric priority into a readable label
SELECT issue_type, priority,
    CASE
         WHEN priority = 3 THEN 'High'
         WHEN priority = 2 THEN 'Medium'
         WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM exercise.class.tickets;
```

Add parameter

Table

	issue_type	priority	priority_label
1	Login issue	1	Low
2	Server down	3	High
3	Slow system	2	Medium
4	Email error	2	Medium
5	Password reset	1	Low

```sql

--Question 07: Calculate attendance percentage and classify the result.
SELECT
    student_id,
    (days_present/total_days)*100 AS attendance_percentage,
    CASE
        WHEN attendance_percentage >= 90 THEN 'Excellent'
        WHEN attendance_percentage BETWEEN 75 AND 89 THEN 'Good'
        WHEN attendance_percentage < 75 THEN 'Needs Improvement'

    END AS attendance_status
FROM exercise.class.attendance;
```

Add parameter

Table

	student_id	attendance_percentage	attendance_status
1	1	90	Excellent
2	2	60	Needs Improvement
3	3	96	Excellent
4	4	50	Needs Improvement
5	5	100	Excellent

```sql
--Question 08: Lable the stock status of each product
SELECT product_id, stock_qty,
    CASE
        WHEN stock_qty = 0 THEN 'Out of stock'
        WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low stock'
        WHEN stock_qty > 5 THEN 'In stock'
END AS stock_status
FROM exercise.class.products_inventory;
```

Add parameter

Table

	product_id	stock_qty	stock_status
1	1	5	Low stock
2	2	0	Out of stock
3	3	25	In stock
4	4	10	In stock
5	5	3	Low stock

```sql
--Question 09: Classify each by the number of enrolled students
SELECT subject, enrolled_students,
    CASE
        WHEN enrolled_students >= 25 THEN 'Large'
        WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
        ELSE 'Small'
        END AS class_size_category
FROM exercise.class.classes;
```

Add parameter

Table

	subject	enrolled_students	class_size_category
1	Math	30	Large
2	English	25	Large
3	Science	15	Medium
4	Art	5	Small
5	History	20	Medium

```sql
FROM exercise.class.payments;

--Question 10: Apply a discount flag based on the payment method and amount
SELECT payment_id, payment_method, amount,
    CASE
        WHEN payment_method = 'Cash' AND amount >= 200 THEN 'Eligible for Discount'
        ELSE 'Not Eligible'
    END AS discount_eligibility
    FROM exercise.class.payments;
```

Add parameter

Table

	payment_id	payment_method	amount	discount_eligibility
1	1	Card	50	Not Eligible
2	2	Cash	200	Eligible for Discount
3	3	Card	150	Not Eligible
4	4	PayPal	75	Not Eligible
5	5	Cash	300	Eligible for Discount